



Mathematics Year 8
Investigation 2 2014

Friday 2nd May

Solution

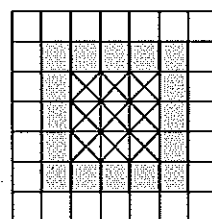
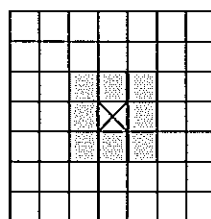
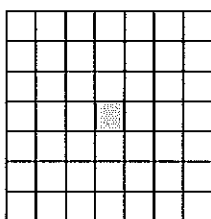
- Search Pattern-

Marks

/10 marks

This is an IN CLASS investigation. You must investigate the problem and write your report in the class room in the time your teacher has allocated.

A hiker has become lost in the bush and the local state emergency service commander has to find them. In order to search in a logical pattern the commander has divided the countryside into 1 kilometre. During the first day, the small search team searches one of the squares (the last known whereabouts of the hiker).



On the second day, the commander has obtained more searchers and searches the eight squares surrounding the initial square (now crossed out). Each day the commander finds more searchers and searches the squares surrounding the previous day's areas. This will continue until the hiker is (hopefully) found. Your job is to help the commander determine the number of squares that need to be searched each day.

SOME HELPFUL IDEAS:

- As usual it is best to work from the easier days up to hard days
- Put your results into a table.
- On the square paper, draw diagrams of the next two days to help you see the pattern.
- Try to look for rules or patterns that can save you some time and effort.
- When you are writing your report it is a good idea to include an introduction and conclusion. The introduction tells the reader about the problem so that they can understand your answers. The conclusion summarises what you found into one paragraph. Because this is an in-class investigation, your introduction and conclusion should be very short.

a) Introduction : (What are you trying to do ?) [2 mark]

Your job is to help the commander determine the number of squares that need to be searched each day.

Use of tables and/or diagrams to help you work out the information.

- b) Work out the correct number of squares to be searched for days 1 through to 10.
Fill out the table after you have figured out the pattern.
Use the square grid paper provided to draw diagrams to help you solve the pattern.
[4 marks]

DAY	NUMBER OF SQUARES
1	1
2	8
3	16
4	24
5	32
6	40
7	48
8	56
9	64
10	72

(1 mark for each 2 answers correct)

- c) Work out the TOTAL number of squares to be searched over the full ten days. [1 mark]

$$= 361$$

- d) Find and EXPLAIN the rules and patterns that you found. [2 marks]

2 marks → The Rule is $8n - 8 = (8 \times \text{DAY No.}) - 8$

1 mark if they mention a difference of 8 is added to successive days.

- e) Use your rule to predict the number of squares to be searched on day 26 if the search went on that long.

[1 mark]

$$= 8n - 8$$

$$= 8 \times 26 - 8$$

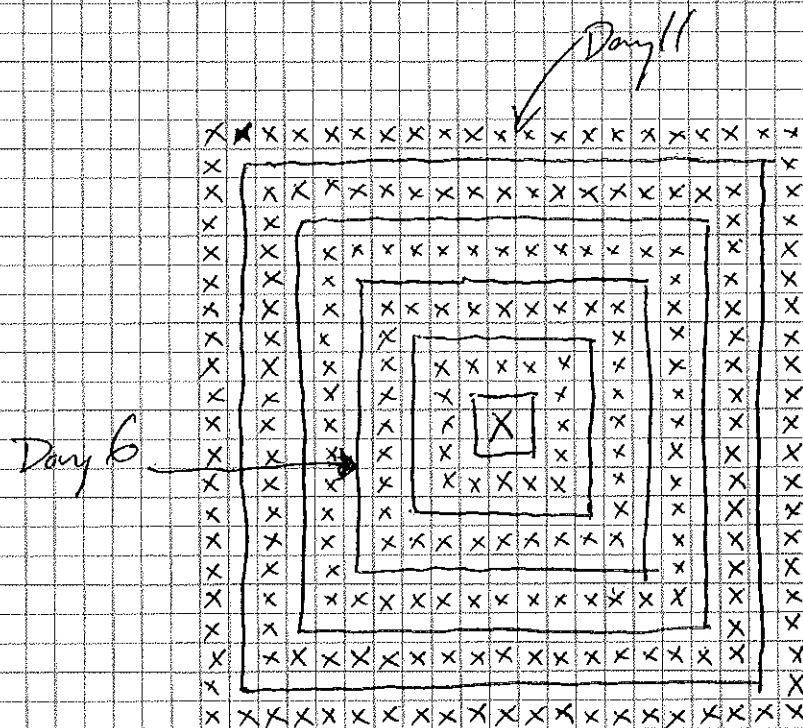
$$= 200$$

DIAGRAM SOLUTION

Day 1 in Middle. (X)

Day 3, 5, 7, 9, 11 are (X).

Day 2, 4, 6, 8, 10 are (—) line



Day	Boxes.
11	80
10	72
9	64
8	56
7	48
6	40
5	32
4	24
3	16
2	8
1	1

DAY	1	2	3	4	5	6	7	8	9	10
No. SQUARES	1	8	16	24	32	40	48	56	64	72
Difference between Days.		3+3 1+1	5+5 3+3	7+7 5+5	9+9 7+7	11+11 9+9	13+13 11+11	15+15 13+13	17+17 15+15	19+19 17+17
		7	8	8	8	8	8	8	8	8